

```
1  /*
2     यदा यदा हि धर्मस्य स्तान्निवृति भारत।
3     अभ्युत्थानमधर्मस्य तदात्मानं सृजाम्यहम्॥
4  */
5  /*
6     ॐ त्र्यम्बकं यजामहे सुगन्धिं पुष्टिवर्धनम् ।
7     उर्वारुकमिव बन्धनान्मृत्योर्मुक्षीय मामृतात्॥
8  */
9
10 #include <bits/stdc++.h>
11 using namespace std;
12
13 int binarySearch(vector<int> &v1, int lo, int hi, int target) {
14     if (lo > hi) {
15         return -1;
16     }
17     int mid = lo + ((hi - lo) >> 1);
18     if (v1[mid] == target) {
19         return mid;
20     }
21     if (v1[mid] > target) {
22         return binarySearch(&v1, lo, hi:mid - 1, target);
23     }
24 }
```

```
D:\My_Programs\GitHub_Repository\CPP\CodeChef\DAA\Lab_2.cpp
Enter number of elements:5
Enter elements in the Vector:5 1 2 3 6
Enter element to search:3
Element found
Process finished with exit code 0
```

```
13 int binarySearch(vector<int> &v1, int lo, int hi, int target) {
23     }
24     return binarySearch(&v1, lo:mid + 1, hi, target);
25 }
26
27 int main() {
28     int n, tar;
29     cout << "Enter number of elements: ";
30     cin >> n;
31     vector<int> v1(n);
32     cout << "Enter elements in the Vector: ";
33     for (int i = 0; i < n; i++) {
34         cin >> v1[i];
35     }
36     sort(first: v1.begin(), last: v1.end());
37     cout << "Enter element to search: ";
38     cin >> tar;
39     int res = binarySearch(&v1, lo:0, hi:n - 1, target:tar);
40     if (res != -1) {
41         cout << "Element found" << endl;
42     } else {
43         cout << "Element not found" << endl;
44     }
45     return 0;
46 }
47 }
```

```
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Enter number of elements:5
Enter elements in the Vector:5 1 2 3 6
Enter element to search:3
Element found
Process finished with exit code 0
```

```
44     }
45     return 0;
46 }
47 }
```